LESSON 6.7

USING PROPORTIONS TO SOLVE PROBLEMS

LESSON INTRODUCTION

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

LEARNING TARGETS

- I can solve problems using proportions.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.

ESSENTIAL QUESTIONS

- How can proportional reasoning help me find unknowns in everyday life?
- How does scaling ratios help when finding missing parts of equivalent ratios?

LESSON NARRATIVE

This lesson extends previous learning to focus on two specific ways to solve a proportion (multiplying both parts of a ratio by the same factor or solving the equation for the unknown value). Students then examine various students' work that highlight specific misconceptions/errors students make before they try solving problems using proportions on their own.

COMPONENT SUMMARY

6.7.1 WARM-UP *(~5 MIN.)*

Using proportions for a recipe

6.7.2 GUIDED INSTRUCTION *(10 MIN.)*

Comparing two methods for solving proportions

6.7.3 COLLABORATION *(15-20 MIN.)*

Small group error analysis with four examples of misconceptions

6.7.4 BRING IT TOGETHER *(5 MIN.)*

Checkpoint for student reflection

6.7.5 YOUR TURN! *(5-10 MIN.)*

Solving three problems, creating and then solving proportions with no rounding

6.7.6 WRAP-UP *(~ 5 MIN.)*

Reflection on a number line

RESOURCES

GLOSSARY

Key Terms:

- Scale factor

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- 6.7.3 Collaboration could be done as a Gallery Walk. Consider copying or printing the prompts from the four student problems onto chart paper. For smaller groups, make two copies of each poster for a total of eight posters for approximately three to four students per poster.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may look for the “correct” or “right way” to set up proportions and then incorrectly identify other methods of proportions as “incorrect.”
- Students may try to add or subtract instead of multiply or divide when solving proportions.
- Students may multiply numerator times numerator and denominator times denominator rather than correctly solving a proportion.

THINGS TO CONSIDER

- Instruction should not include cross-multiplying as a strategy. Cross-multiplication does not build students’ understanding and often creates a misconception for multiplying fractions.

LITERACY AND LANGUAGE ROUTINES

Clarify, Critique, and Connect

Activities 6.7.2 Guided Instruction and 6.7.3 Collaboration allow students the opportunity to correct their errors and clarify their meaning in the context of others to strengthen their problem-solving and calculations. Use these activities as a space where you facilitate the discussion to uncover student thinking.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

6.7.3 Collaboration provides an opportunity for students to engage in mathematical discussion by analyzing and critiquing the thinking of others. Note the emphasis in process over product as students analyze the thinking and work of others more than the answer.

DIFFERENTIATE INSTRUCTION

Consider a Gallery Walk for 6.7.3 Collaboration. Students can use strategies like Mark-Up for visual scaffolding in identifying key misconceptions or displayed student work. Foster discussion with specific prompts using the questions on each poster in the activity or with specific roles within each group. For example, one student can be the scribe, tasked with taking notes on what students are learning at each poster, while another student is the host, tasked with making sure student talk is equitable. The posters around the room provide opportunities for movement.

6.7.1 WARM-UP: MAKING MORE SERVINGS

TEACHER MOVES

Consider this spiraled practice and not additional instruction. Monitor for student work both to monitor progress from Lesson 6 and to use as observational data for small groups in 6.7.3 Collaboration.

6.7.1 Warm-Up: Making More Servings

1. Gabriel wants to make 5 servings of oatmeal for breakfast using this recipe. How much water and oats will he need to use? Show your work.

TRADITIONAL STEEL CUT OATS
Hot Water or Milk Preparation

1 Serving

- 1-1/2 cups water or milk $\times 5$
- 1/4 cup Quaker Steel Cut Oats $\times 5$

Directions:

1. Bring water or milk to a boil in a medium saucepan.
2. Stir in oats, reduce heat to low.
3. Simmer uncovered over low heat, stirring occasionally, for 25-30 minutes or until oats are of desired texture.

7.5 *cups of water*
1.25 *cups of oats*

6.7.2 Guided Instruction: Solving Proportions

Just as there are different ways to set up a proportion to solve a problem, there are different ways to solve the proportion once it's set up.

Jacksonville and Gainesville are approximately 1.4 inches (in.) apart on a scaled map of Florida. The scale used for the map is 2 in. $=$ 90 miles. The proportion below can be used to find the actual distance between Jacksonville and Gainesville.

$$\frac{2}{90} = \frac{1.4}{x}$$

Two methods for solving the proportion are shown.

Method A **Using a Scale Factor**	**Method B** **Solving the Equation**
$\frac{2}{90} = \frac{1.4}{x}$ Determine the factor multiplied by 2 to get 1.4. $1.4 \div 2 = 0.7$ $\frac{2}{90} \times \frac{0.7}{0.7} = \frac{1.4}{63}$	$\frac{2}{90} = \frac{1.4}{x}$ $\cancel{90} \cdot \frac{2}{\cancel{90}} = \frac{1.4}{x} \cdot 90$ $x \cdot 2 = \frac{126}{\cancel{x}} \cdot \cancel{x}$ $\frac{\cancel{2}x}{\cancel{2}} = \frac{126}{2}$ $x = 63$
Both solution methods show the approximate distance between Jacksonville and Gainesville is 63 miles.	

1. In Method A, how does multiplying the ratio $\frac{2}{90}$ by $\frac{0.7}{0.7}$ show that $\frac{2}{90}$ and $\frac{1.4}{63}$ are equivalent?

   When both the numerator and denominator are multiplied by the same value, the value remains equivalent.

2. In Method B, why were both ratios multiplied by 90 and then by x?
   To balance the equation and maintain equivalence.

3. In Method B, why are both sides of the equation multiplied or divided by the same value in each step?
   To balance the equation and maintain equivalence.

A certain shade of light blue paint is made by mixing $2\frac{1}{2}$ quarts of blue paint with 7 quarts of white paint. To make the same shade of light blue, how much white paint would be needed to mix with 4 quarts of blue paint?

4. Complete the proportion using the information given.

$$\frac{2.5}{4} = \frac{7}{x}$$

5. With your partner, decide who is Partner A and who is Partner B. Then solve the proportion using the method indicated in the space provided.

Partner A - Method A Using a Scale Factor	**Partner B - Method B** Solving the Equation
$\frac{2.5}{4} = \frac{7}{x}$ $7 \div 2.5 = 2.8$ $4 \times 2.8 = 11.2$ *qt.*	$\cancel{4\times}\frac{2.5}{4} = \frac{7}{x} \times 4$ $x \times 2.5 = \frac{2.8}{\cancel{x}} \times \cancel{x}$ $\frac{\cancel{2.5}x}{\cancel{2.5}} = \frac{28}{2.5}$ $\quad x = 11.2$ *qt.*

6. Check your work with your partner, making any necessary corrections. Once both partners agree, write your partner's method in your workbook. Then complete the statement.

   To make the same shade of light blue, 11.2 quarts of white paint would be needed.

6.7.2 GUIDED INSTRUCTION: SOLVING PROPORTIONS

ENRICHMENT

How is this the same as what you did in the previous lesson? How is it different? (Both methods got the same answer.)

Use these questions to continue highlighting for students the flexible pathways for constructing and solving proportions.

QUESTIONING

Consider prompting students to find the distance between their hometown and the city of a family member using a map. This can be a project-based extension at the end of the unit or homework.

DIFFERENTIATE INSTRUCTION

Clarify, Critique, and Connect

Consider constructing questions 4-6 as a language routine, with the teacher acting as the facilitator critique. Students complete question 4 before clarifying with the whole group. Students then complete question 5 before critiquing and connecting their methods in question 6. The teacher should monitor student responses during question 6 to surface accurate mathematical thinking or misconceptions whole group.

6.7.3 COLLABORATION: EXAMINING STUDENT WORK

DIFFERENTIATE INSTRUCTION

Consider a variety of methods for facilitating collaboration based on your students and classroom.

Gallery Walk

- Round 1: Have students answer only the first question about what the student did correctly. Give them five minutes to discuss and write their answer on the poster.
- Round 2: Groups move to the next poster to answer the second question within five minutes.
- Round 3: Groups move to the next poster to answer the third question within five minutes.
- Round 4: Groups move to their final poster to review and revise the thoughts of all the other groups for all three questions.

TEACHER MOVES

Teacher Note for Javier's Work

Students should be noting the difference between part: part (5:3) and part: whole (5:8), with Javier incorrectly using 160 as part instead of whole.

6.7.3 Collaboration: Examining Student Work

Four students solved different problems involving proportional relationships. Each student's work is shown, including any errors they may have made. Work with your partner or a small group to examine each student's work and complete the table that follows.

Student	Javier
Problem	A mixture of concrete is made up of sand and cement in a ratio of $5:3$. How many cubic feet of sand is needed to make 160 cubic feet of concrete?
Student's Work	$\times 32$ $\frac{5}{3} = \frac{160}{x}$ $\times 32$ $x = 96\ ft^3$

What did Javier do correctly?	What error(s) did Javier make?	Find the correct answer.
He solved the proportion correctly.	*He set up the proportion wrong. He used sand to cement instead of sand to concrete (total parts).*	$x\ 20$ $\frac{5}{8} = \frac{x}{160}$ $x\ 20$ $\boxed{100\ ft^3}$

Student	Kayla
Problem	A recipe for making 4 pancakes calls for the ingredients listed. ☐ 6 tablespoons flour ☐ $\frac{1}{4}$ pint milk ☐ 1 pinch salt ☐ 1 egg ☐ $\frac{1}{3}$ pint water You want to make 10 pancakes. a. How much flour do you need? b. How much milk do you need?
Student's Work	a. How much flour do you need? 10 : 4 = ? : 6 Flour • $\frac{10}{4} = \frac{?}{6}$ 10 pancakes = 6 + 6 + 3 = 15 spoonfuls of flour b. How much milk do you need? 10 pancakes = $\frac{1}{4} + \frac{1}{4} + \frac{1}{8} = \frac{3}{16}$ pints

What did Kayla do correctly?	What error(s) did Kayla make?	Find the correct answers.
• *She found the right amount of flour needed.* • *She used a strategy that would work by relating what is needed for 4 to 10 pancakes.*	• *She used the wrong units for flour (she used spoonfuls instead of tablespoons).* • *She added fractions incorrectly. (Find common denominator first.)*	*15 Tbsp* $\frac{2}{8} + \frac{2}{8} + \frac{1}{8}$ $= \frac{5}{8}$ *pt. milk*

6.7.3 COLLABORATION: EXAMINING STUDENT WORK (CONTINUED)

TEACHER MOVES

Teacher Note for Kayla's Work

Students may note that she set up the proportion "incorrectly." Prompt students to recognize the variety of methods from the previous lesson and how different methods can still produce the same result.

Examples:

$\frac{\text{pancake (10)}}{\text{pancake (4)}} = \frac{\text{flour (x)}}{\text{flour (6)}}$ or $\frac{\text{pancake (4)}}{\text{flour (6)}} = \frac{\text{pancake (10)}}{\text{flour (x)}}$

look different but produce the same result.

Student	Isaac
Problem	A drawing of a surfboard in a catalog shows its length as $8\frac{4}{9}$ inches (in.). Find the actual length of the surfboard if $\frac{1}{2}$ in. length on the drawing corresponds to $\frac{3}{8}$ foot of actual length.
Student's Work	$\frac{\frac{1}{2}}{8\frac{4}{9}} = \frac{\frac{3}{8}}{s}$ $8.44 \cdot \frac{0.5}{8.44} = \frac{0.38}{s} \cdot 8.44$ $s \cdot 0.5 = \frac{3.2072}{s} \cdot s$ $\frac{0.5s}{0.5} = \frac{3.2072}{0.5}$ s = 6.4 in.

What did Isaac do correctly?	What error(s) did Isaac make?	Find the correct answers.
• *He set up the proportion correctly.* • *He used a correct strategy to solve it.*	• *He rounded the fractions, so he ended up with an inaccurate answer.*	$\frac{\frac{1}{2}}{8\frac{4}{9}} = \frac{\frac{3}{8}}{s}$ $s = \frac{\frac{76}{9} \times \frac{3}{8}}{\frac{1}{2}}$ $s = \frac{\frac{228}{72}}{\frac{1}{2}} = \frac{228}{72} \times \frac{2}{1}$ $s = \frac{456}{72} = \frac{19}{3}$ $s = 6\frac{1}{3}$

TEACHER MOVES

Teacher Note for Isaac's Work

Fluency with fractions will be a helpful skill in completing Isaac's work.

6.7.3 COLLABORATION: EXAMINING STUDENT WORK (CONTINUED)

TEACHER MOVES

Teacher Note for Mari's Work

Students may wonder what finding the "smallest number" of shots to have a better percentage than Juan means.

Students can either identify that the proportion should have 14 shots instead of 13 shots OR that the final answer should be rounded up (as seen in the key).

Student	Mari	
Problem	Juan made 13 out of 20 free throws at basketball practice. If Jada shoots 25 free throws, what is the smallest number of shots she has to make to have a better free-throw percentage than Juan?	
Student's Work	$\frac{13}{20} = \frac{f}{25}$ $20 + 5 = 25$ $13 + 5 = 18$ Jada has to make 18 shots.	
What did Mari do correctly?	**What error(s) did Mari make?**	**Find the correct answers.**
She set up the proportion correctly.	*She added instead of multiplying to solve the equation.*	$\frac{13}{20} \overset{\times 1.25}{=} \frac{f}{25}$ (× 1.25) 16.25 *She would have to make 17 shots.*

6.7.4 BRING IT TOGETHER: PROPORTIONS

DIFFERENTIATE INSTRUCTION

Utilize moments of reflection like this to cultivate growth mindset and self-regulation. Consider using sentence stems from growth mindset tools, MA.K12.MTR.1.1, or Universal Design for Learning Guidelines.

- Promote expectations and beliefs that optimize motivation (partner shout-outs).
- Facilitate personal coping skills and strategies (celebrate growth as much as or more than you celebrate proficiency).
- Develop self-assessment and reflection (recognize specific reflections that include examples).

6.7.4 Bring It Together: Proportions

1. Mistakes create opportunities to learn! Based on the mistakes you observed when examining other students' work, use the space provided to describe at least two things you learned and want to keep in mind when using proportions to solve problems.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Answers will vary.

Ensure the proportion is set up correctly prior to solving.

Ensure that your fraction is correct when using the scale factor to solve the proportion.

6.7.5 Your Turn!

Solve each problem using proportional reasoning. Show your work.

1. Tyler swims at a constant speed of 5 meters (m) every 3 seconds. How long does it take him to swim 114 meters?

   $\frac{5\ m}{3\ sec} = \frac{114\ m.}{x}$ $\boxed{x = 68.4\ sec.}$

   OR

   $114\ \cancel{m.} \times \frac{3\ sec.}{5\ \cancel{m.}}$

2. To make a shade of paint called jasper green, mix 4 quarts (qt) of green paint with $\frac{2}{3}$ cup (c) of black paint. How much green paint should be mixed with 4.5 cups of black paint to make jasper green?

   $\frac{green}{black} \Big| \frac{4}{\frac{2}{3}} = \frac{?}{4.5}$ $\boxed{27\ quarts}$

3. It takes an ant farm 3 days to consume $\frac{1}{2}$ of an apple. At that rate, in how many days will the ant farm consume 6 apples?

   $\frac{3}{\frac{1}{2}} = \frac{x}{6}$ $\boxed{36\ days}$

6.7.5 YOUR TURN!

ENRICHMENT

What similarities and differences do you see between the ways that you solved these problems? Why did you decide to structure your proportions in these ways?

6.7.6 Wrap-Up: Reflection

1. Draw an **X** on each line to indicate where your current learning is in relation to today's learning target.
   Answers will vary.

   I can solve problems using proportions.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this, and I feel confident.

2. Explain why you placed your **X** where you did.
   Answers will vary.

6.7.6 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

To help students break down where their strengths and weaknesses lie, consider including multiple scales with the following prompts.

- I can set up a proportion correctly.
- I can solve a proportion correctly.
- I can determine if and when an answer should be rounded.

These results can also help you identify class trends for reteaching and remediation.

